Applied Machine Learning (CSAI2017P)
Test-0 — Diagnostic

Duration 30 minutes • **This test carries no marks**

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Name:	____________	**SAP ID:**	____________
Batch:	____________	**Date:**	____________

Read this first

This test carries no marks and will not affect your grade in any way.
Its purpose is to let me see how you think about a problem, so that I can pitch the lectures and the worked examples where they are actually needed. It does not sort you into anything, and it has no bearing on which version of any later test you are given — those are handed out at random.
There is **no penalty** for an incomplete answer that shows its reasoning, and **no credit** for a bare correct answer with no working.
If you do not know something, say so, and say what you would need to find out. That is a useful answer, not a blank one.

Instructions. Individual, closed book. Write by hand in the space provided; continue on the back of the sheet if you need to. Do not erase your working — crossings-out are useful to me. Attempt every block; a partial attempt is better than a blank.

Block A — Warm-up (5 items, about 4 minutes)

Circle one answer. No working needed for this block.

A1. A fair coin is tossed twice. The probability of getting exactly one head is:
(a) 1/4 (b) 1/2 (c) 3/4 (d) 1

A2. In Python, `len({'a': 1, 'b': 2})` evaluates to:
(a) 1 (b) 2 (c) 4 (d) an error

A3. A is 3×4 and v is a column vector. For Av to be defined, v must have shape:
(a) 3×1 (b) 4×1 (c) 1×3 (d) 1×4

A4. For the values $2, 3, 4, 5, 96$, which is larger?
(a) the mean (b) the median (c) they are equal (d) cannot be determined

A5. A *training set* is:
(a) data used to fit the model (b) data used to report final performance
(c) data with no labels (d) a subset of the test set

Block B — Working

(3 items, about 8 minutes)

Show your working. There is more than one valid route through each of these; I am interested in the route you take.

B1. A factory has two machines. The first makes 60% of the items and 2% of what it makes is defective. The second makes the rest, and 5% of what it makes is defective. An item is picked at random and found to be defective. How likely is it that the first machine made it?

B2. For a fixed set of numbers $y_1, \ldots, y_n$, find the value of c that makes $S(c) = \sum_{i=1}^{n}(y_i - c)^2$ as small as possible. State how you know your answer is a minimum and not a maximum.

B3. Consider the following function.

```python
def summarise(values):
    total = 0
    for v in values:
        total = total + v
    return total / len(values)
```

What does `summarise([2, 4, 6])` return? What does `summarise([])` do? Explain how you worked each one out.

Block C — Explain it

(1 item, about 5 minutes)

C1. In at most 120 words, explain to a first-year student what it means for a model to **overfit**, and give one concrete example. Use whatever form of explanation you find clearest — words, a sketch, a small table, a formula, an analogy.

Block D — Debug

(1 item, about 6 minutes)

D1. The function below is meant to train and evaluate a model honestly. There is exactly one fault in it.

```
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier

def evaluate(X, y):
    scaler = StandardScaler()
    X_scaled = scaler.fit_transform(X)

    X_train, X_test, y_train, y_test = train_test_split(
        X_scaled, y, test_size=0.25, random_state=0)

    model = KNeighborsClassifier(n_neighbors=5)
    model.fit(X_train, y_train)
    return model.score(X_test, y_test)
```

Answer three things: **(i)** what is wrong, **(ii)** why it matters, and **(iii)** how you would confirm that this is really the problem.

Block E — Getting started

(1 item, about 5 minutes)

E1. A shop wants you to predict whether a customer will return next month. Describe how you would start.

Block F — Estimation

(1 item, about 2 minutes)

F1. Roughly how many numbers are stored in a 1000×1000 greyscale image, and roughly how much memory would 500 such images need? Show your reasoning and state any assumption you make.

End of Test-0. Hand in this sheet whether or not it is complete — a partial script is still useful.
Thank you; the tests you sit later this semester will be shaped by what I read here.